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A 3B C12

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Subject: Programming fundamentals

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Section: A

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10A

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Task 1

Program 1

```
#include <iostream>
unsing namespace std;
int main( ){
Char letter;
letter = 'A';
Cout << letter << "\n";
letter = 'B';
Cout << letter << "\n";
return 0;
}
```

out Put | A , B |

Program 2

```
# include < iostream>
using namespace std;
int main( ){
```

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```
char letter;
 letter = 65;
cout << letter << "\n";
 letter= 66;
cout << letter << "\n";
 return 0;
}
```

outPut [A, B]

Program 3

```
#include <iostream>
using namespace std;
int main(){
int letter;
 letter = 'A';
cout<<letter << "\n";
 letter = 'B';
cout << letter << "\n";
 return 0;
}
```

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out Put : 65 , 66

Task 2

```
# include <iostream>
using namespace std;
int main( ) {
int num1 , num2 ;
cout << " Enter First number : " ;
 cin >> num1 ;
cout << " Enter second number : " ;
cin >> num2 ;
cout << "Addition : " << num1 + num2 << endl ;
cout << " Subtraction : " << num1 - num2 << endl ;
cout << " Multiplication : " << num1 * num2 << endl ;
Cout << " Division : " << num1 / num2 << endl ;
cout << " Modulus : " << num1 % num2 << endl ;
 return 0 ;
 }
```

Task 3

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Task 3

```
int val1, val2, val3;
cout << "Please enter your 3 numbers:";
cin >> val1 >> val2 >> val3;
cout << val1 << "\n" << val2 << "\n" <<
    val3 << "\n";
cout << val3 << "\n" << val2 << "\n" << val1
    << "\n";
```

if user enters : 10, 20, 30

output will be

10
20
30
30
20
10

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Task 4

```
# include <iostream>
# include <cmath>
using namespace std;
int main() {
double x1 , Y1 , x2 , Y2 , distance;
cout << "Enter x1 , Y1 : " ;
cin >> x1 >> Y1 ;
Cout << "Enter X2 , Y2 : " ;
cin >> x2 >> Y2 ;
distance = sqrt ( Pow(x2-x1 , 2) + Pow(
  Y2 - x1 , 2) ) ;
cout << "Distance = " << distance ;
  return 0 ;
  }
```

Task 5

```
# include <iostream>
using namespace std;
int main ( ){
// 1. circle
double r , aredCircle ;
```

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```
const double PI = 3.14159;
cout << "Enter radius of Circle:";
  cin >> r;
areaCircle = PI * r * r;
cout << " Area of Circle = " << areaCircle << endl;

// 2. Rectangle
double length, width, areaRect;
cout << " Enter length and width of Rectangle: ";
cin >> length >> width;
areaRect = length * width;
cout << " Area of Rectangle = " << areaRect << endl;

// 3. Triangle
double base, height, areaTri;
cout << " Enter base and height of Triangle: ";
cin >> base >> height;
areaTri = base * height * 0.5;
cout << " Area of Triangle = " << areaTri << endl;
  return 0;
  }
```

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Task 6

```
# include < iostream >
using namespace std;
int main( ){
double item1 = 12.95, item 2 = 24.95, item3 = 6.95,
item4 = 14.95, item5 = 3.95;
double subtotal, salesTax, total;
subtotal = item1 + item2 + item3 + item4 + item5 ;
salesTax = subtotal * 0.06 ; // 6%
total = subtotal + salesTax ;
cout << "Price of item 1 = $ " << item1 << endL;
cout << "Price of item2 = $" << item2 << endL;
cout << "Price of item 3 = $" << item3 << endL;
cout << "Price of item4 = $" << item4 << endL;
Cout << "Price of item5 = $" << item5 << endL;
cout << " subtotal = $ " << subtotal << endL;
cout << " sales Tax (6%) = $ " << salesTax << endL;
cout << " Total = $ " << total << endL;
  return 0;
}
```

Bonus Task

```
# include < iostream >
using namespace std;
```

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```
int main ( ) {
  int num;
cout << " Enter a five-digit integer;
  cin >> num;
int d1 = num / 10000;
int d2 = (num / 1000) % 10;
int d3 = ( num / 100 ) % 10;
int d4 = ( num / 10 ) % 10;
int d5 = num % 10;
cout << d1 << " " << d2 << " " << d3 << " "
  << d4 << " " << d5;
  return 0;
  }
```

if you input 1, 2, 3, 4 5 output will be:

out put : 1 , 2 3 4 5

Home Task 1

```
# include < iostream >
using namespace std;
 int main ( ){
int var 1 , var 2 , temP ;
Cout << " Enter var 1 : " ;
```

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```
cin >> var1;
cout << " Enter var2 : ";
cin >> var2;
cout << "\nBefore swap: var1 = " << var1 << ", var2
= " << var2 << endl;
// swapping logic
temp = var1;
var1 = var2;
var2 = temp;
cout << " After swap: var1=" << var1 << "\nvar2=
" << var2 << endl;
return 0;
}
```

inPut = var1 = 10 , var2 = 20
outPut = var1 = 20, var 2 = 10

Home Task 2

```
# include <iostream>
using namespace std;
int main ( ) {
char small, capital;
cout << "Enter alphabet in small letter (a to z): ";
```

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```
cin >> small;
// method 1 - using ASCII
capital = small - 32;
cout << "capital letter: " << capital << endL;
// Method 2 - using built-in function (extra)
// capital - toupper (small);
return 0;
}
```

InPut: b
outPut: B

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